

Autodesk® Scaleform®

S c a l e f o r m I n t e g r a t i o n T u t o r i a l

D i r e c t X 9 W i n d o w s S c a l e f o r m 3 D

: Ben Mowery
: 3.09
: 2013 10 7

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Autodesk® Scaleform® 4.3

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M E R C H A N T A B I L I T Y O R F I T N E S S F O R A P A R T I C U L A R P U R P O S E R E G A R D I N G T H E S E
M A T E R I A L S .

A u t o d e s k S c a l e f o r m :

S c a l e f o r m 4 . 2 (S c a l e f o r m 4 . 2 I n t e g r a t i o n T u t o r i a l)

S c a l e f o r m C o r p o r a t i o n

6 3 0 5 I v y L a n e , S u i t e 3 1 0

G r e e n b e l t , M D 2 0 7 7 0 , U S A

[w w w . s c a l e f o r m . c o m](http://www.scaleform.com)

[i n f o @ s c a l e f o r m . c o m](mailto:info@scaleform.com)

(3 0 1) 4 4 6 - 3 2 0 0

F A X

(3 0 1) 4 4 6 - 3 1 9 9

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1 はじめに

注意: `ScaleForm` `ScaleForm` `ScaleForm` `ScaleForm`

注意: Scale form

Scale form 4

2 ドキュメントの概要

Scaleform Developer Center
(<https://developer.scaleform.com/gfx>)

:

- X Web Scaleform SDK
- X PDF
- X Font Overview :
Scaleform C++ API
- X XML Overview : Scaleform XML
- X Scale9Grid : Scale9Grid
- X IME Configuration : Scaleform IME
Flash IME
- X ActionScript Extensions : Scaleform ActionScript

3 インストールとビルドの依存関係

3.1 インストール



注意:

. s i n V i s u a l S t u d i o

W i n 3 2

I n c l u d e

S c a l e f o r m

L i b

S c a l e f o r m

R e s o u r c e s

C L I K

D o c

2

P D F

3.2 デモのコンパイル

S c a l e f o r m

C : \ P r o g r a m F i l e s (x 8 6) \ S c a l e f o r m \ G F x S D K 4 . 3 \ P r o j e c t s \ W i n 3 2 \ M s v c 9 0 \ D e m o s \ G F x 4 . 1

D e m o s . s i n

V i s u a l S t u d i o

. s i n

D 3 D 9 _ D e b u g _ S t a t i c

S c a l e f o r m P l a y e r

C : \ P r o g r a m F i l e s \ S c a l e f o r m \ G F x S D K

4 . 3 \ B i n \ W i n 3 2 \ M s v c 9 0 \ G F x P l a y e r \ G F x P l a y e r _ D 3 D 9 _ D e b u g _ S t a t i c . e x e

[] Æ

] Æ

[S c a l e f o r m] Æ

[G F x S D K 4 . 3] Æ

[D e m o E x a m p l e s]

G F x P l a y e r D 3 D x

S W F

S c a l e f o r m F l a s h

3.3 Scaleform Player における SWF 再生のベンチマーク

[] Æ

] Æ

[S c a l e f o r m] Æ

[G F x S D K 4 . 3] Æ

[D e m o E x a m p l e s] Æ

[G F x P l a y e r D 3 D 9]

C : \ P r o g r a m F i l e s \ S c a l e f o r m \ G F x S D K

4 . 3 \ B i n \ D a t a \ A S 2 \ S a m p l e s \ S W F t o T e x t u r e

3 D W i n d o w . s w f



1:

F l a s h

F 1

1. C t r l + F

2. ctrl + w

F P S

F l a s h

Scale form

C t r l + W

3.

C t r l

4.

C t r l

5. `Ctrl + Z`

6. `Ctrl + U`

7. F 2

/

F l a s h

1

8 0 0 x 6 0 0

1 6 0 0 x 1 2 0 0

1

" S c a l e f o r m "

Scale form

F l a s h

Scale form

[3 D S c a l e f o r m L o g o] (3 D S c a l e f o r m)

C t r l + W

```

Ctrl+ A                                Edge Anti-Aliasing
(                                     ( Edge A A )
                                     ( F S A A )
                                     F S A A
                                     4
                                     S c a l e f o r m
Edge A A
                                     ( D P )
Edge A A
S c a l e f o r m P l a y e r           F l a s h
                                     C P U
                                     S c a l e f o r m
                                     f p s
                                     C P U
Ctrl+ Y                                ( 6 0 f p s )
C P U

```

注意:

S c a l e f o r m

3.4 サンプル ベースのコンパイル

```
Scaleform 4.3
Program Files Scaleform \GFX SDK 4.3 \Apps \Tutorial Tutorial
Windows [ ] [Scaleform] AE[GFX SDK 4.3] AE[Tutorial]
Visual Studio 2005 Visual Studio 2008
"Debug"
```

図 2. Shadow Volume UI

3.5 Scaleform ビルドの依存関係

```
Scaleform Visual
Studio
$(GFX SDK) SDK
C:\Program
Files\Scaleform\GFX SDK 4.3\; lib $(GFX SDK)
lib "Msvc90"
Visual Studio 2005, Visual Studio 2008 Visual Studio 2010
Msvc80, Msvc90 Msvc10
Scaleform
```

```

$(GFXSDK)\src
$(GFXSDK)\include

VisualStudio [

    Gfx.h

    AS3

    AS3

    Gfx/AS3/AS3_Global.h

    AS3

```

Scaleform LITE Customization

The following library directories should be added to the "Additional Library Directories" field for all build configurations:

```

"%$(DXSDK_DIR)\Lib\x86";

"%$(GFXSDK)\3rdParty\expat-2.1.0\lib\$(PlatformName)\Msvc90\Release";

"%$(GFXSDK)\3rdParty\pcre\Lib\$(PlatformName)\Msvc90\Release";

"%$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release";

"%$(GFXSDK)\3rdParty\libpng-1.5.13\Lib\$(PlatformName)\Msvc90\Release";

"%$(GFXSDK)\3rdParty\curl-7.29.0\lib\$(PlatformName)\Msvc90\Release";

"%$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug";

"%$(GFXSDK)\Lib\$(PlatformName)\Msvc90\$(ConfigurationName)";

```

注意: Msvc90 VisualStudio

Scaleform

```

libgfx.lib

libgfx_as2.lib

libgfx_as3.lib

libgfx_air.lib

libgfxexpat.lib

libjpeg.lib

zlib.lib

libcurl.lib

libpng.lib

libgfxrender_d3d9.lib

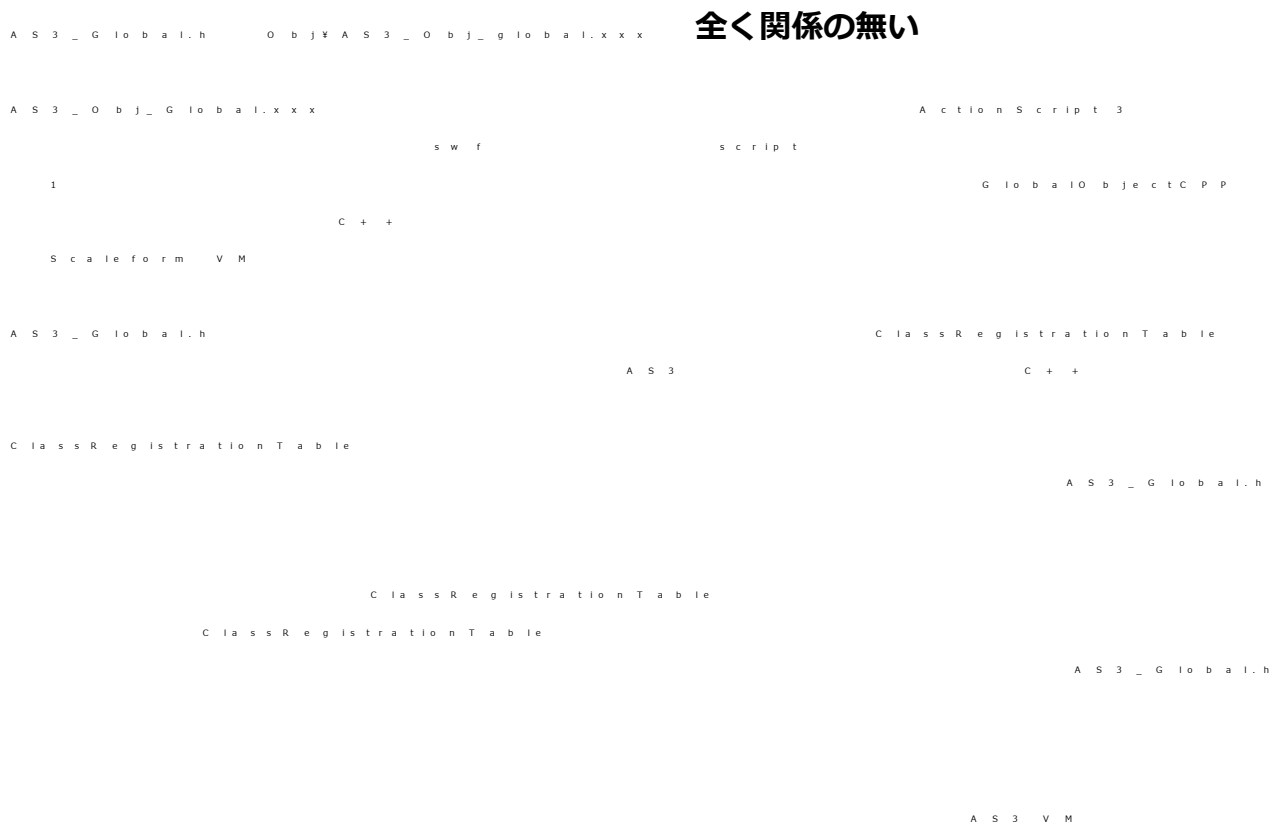
libgfxsound_fm od.lib

```

Scaleform

.vcproj Tutorial Section 3.5

3.5.1AS3_Global.h と Obj¥AS3_Obj_Global.xxx ファイル



4 ゲームエンジンの統合

Scaleform Unreal Engine 3 Gamebryo™ Bigworld Hero Engine™ Touchdown
Jupiter Engine™ CryENGINE™ Trinity Vision Engine™ 3D

Scaleform

DirectX Scaleform
DirectX Shadow Volume SDK DirectX

3D DXUT 2D UI

Scaleform DXUT
Flash Scaleform

DXUT

Win32 DirectX

Scaleform SDK

GfxPlayerTiny.cpp

4.1 Flash のレンダリング

3D Flash

Flash

Gfx::Loader

Flash

Gfx::MovieDef

Gfx::Movie

Render::Renderer2D

Render::HAL

Scaleform

API

DirectX

/

Shadow Volume

Tutorial\Section 4.1

1 :

Shadow Volume.cpp

```

#include "Gfx_Kernel.h"
#include "Gfx.h"
#include "Gfx_Renderer_D3D9.h"

// G F X T u t o r i a l
// S c a l e f o r m
// S c a l e f o r m
// S c a l e f o r m

// G F X : : L o a d e r
Loader          gfxLoader;

// S W F / G F          G F : : M o v i e D e f
Ptr<MovieDef>    pUIMovieDef;

//          G F : : M o v i e
Ptr<Movie>      pUIMovie;

// R e n d e r e r d a t a
Ptr<Render::D3D9::HAL> pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle      hMovieDisplay;

# 2 : G F X : : S y s t e m

S c a l e f o r m          G F X : : S y s t e m
S c a l e f o r m          W i n M a i n

//          G F : : S y s t e m
Gfx::System          gfxInit;

G F X : : S y s t e m          S c a l e f o r m
S c a l e f o r m
W i n M a i n
G F X : : S y s t e m          S c a l e f o r m
G F X : : S y s t e m
G F X : : S y s t e m          G F X T u t o r i a l

```



```

G F x :: S y s t e m

G F x :: S y s t e m :: I n i t ( )      G F x :: S y s t e m :: D e s t r o y ( )

# 3 :

S c a l e f o r m                                W i n M a i n
I n i t A p p ( )                                I n i t A p p ( )

:

gFx = new GFxTutorial();
assert(gFx != NULL);
if(!gFx->InitGFx())
assert(0);

G F x T u t o r i a l      G F x :: L o a d e r
G F x :: L o a d e r      1      S W F / G F x

S W F / G F x

G F x :: L o a d e r      G F x :: L o g

G F x T u t o r i a l :: I n i t G F x ( )      G F x :: L o a d e r
G F x :: L o a d e r      S e t L o g
G F x
S c a l e f o r m      P l a y e r L o g

G F x :: L o g

//      - - G F
//

gfxLoader->SetLog(Ptr<Log>(*new Log()));

G F x :: L o a d e r      G F x :: F i l e O p e n e r
G F x :: F i l e O p e n e r

//
P t r < F i l e O p e n e r >      p f i l e O p e n e r = * n e w      F i l e O p e n e r ;
gfxLoader->SetFileOpener(pfileOpener);

R e n d e r :: H A L      S c a l e f o r m
D 3 D 9
S c a l e f o r m      D 3 D
H A L :: I n i t H A L

```

I D i r e c t 3 D e v i c e 9 R e n d e r H A L S c a l e f o r m
 D X 9 U I

```

pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr()))
    return false;
  
```

S c a l e f o r m R e n d e r : : D 3 D 9 : : H A L
 R e n d e r : : H A L S c a l e f o r m

4 : F l a s h

G F x : : L o a d e r

G F x : : M o v i e D e f

G F x : : M o v i e D e f

G F x : : M o v i e D e f

A c t i o n S c r i p t

```

//
pUIMovieDef = *gFxLoader.CreateMovie(UIMOVIE_FILENAME
                                     Loader::LoadKeepBindData |
                                     Loader::LoadWaitFrame1, 0);
  
```

L o a d K e e p B i n d D a t a

D 3 D

L o a d W a i t F r a m e 1

C r e a t e M o v i e

G F x : : T h r e a d e d T a s k M a n a g e r

M e m o r y S y s t e m O v e r v i e w

5 :

G F x : : M o v i e D e f

G F x : : M o v i e

G F x : : M o v i e

A c t i o n S c r i p t

```

pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
  
```

Create Instance

```
raise ActionScript  
ActionScript
```

Flash

Memory System

Overview

Advance()

raise Create Instance

```
// Advance  
pUIMovie->Advance(0.0f, 0, true);  
  
// time  
//  
MovieLastTime = timeGetTime();
```

Advance

)

時間差 (

3D

:

pUIMovie->SetBackgroundAlpha(0.0f);

Flash

3D

6 :

DirectX

Scale form

Render::D3D9::HALInitParams

Render::D3D9::HAL::InitHAL

D3D

Scale form

/

D3D

InitHAL

DXUT

Shadow Volume OnResetDevice

GfxTutorial

OnResetDevice

:

pRenderHAL->InitHAL(

Render::D3D9::HALInitParams(pd3dDevice, presentParams,

Render::D3D9::HALConfig_NoSceneCalls));

```

InitHAL()          D3D          Scaleform

HAL_NoSceneCalls   DirectX_BeginScene() end_EndScene()

Scaleform          ShadowVolume

OnFrameRender

```

```
# 7 :
```

```

D3D          D3D

ShadowVolume     OnLostDevice

Render::HAL

GfxTutorial     OnLostDevice     D3D

```

pRendererHAL->ShutdownHAL();

```

DXUT          Win32/DirectX

Scaleform_SDK     Scaleform_PlayerTiny.cpp

# 8 :

Scaleform          ScaleformTutorial

WinMain          ScaleformTutorial

```

delete gfx;
gfx = NULL;

```

DirectX_9          Scaleform

InitHAL()          ShutdownnHAL()

DirectX_9          InitHAL

D3DPOOL_DEFAULT   D3DPOOL_DEFAULT   InitHAL

ShutdownnHAL       D3DPOOL_DEFAULT   ShadowVolume

DXUT          D3DPOOL_DEFAULT

DXUT_OnResetDevice     OnLostDevice

GfxTutorial::OnLostDevice     ShutdownnHAL

GfxTutorial::OnResetDevice     InitHAL

```

```

INITIAL SHUTDOWNHAL
D3DPOOL_DEFAULT

# 9 :

D3D
GfxTutorial::OnResetDevice()

//
RECT windowRect = DXUTGetWindowClientRect();
DWORD windowWidth = windowRect.right - windowRect.left;
DWORD windowHeight = windowRect.bottom - windowRect.top;
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
                      windowHeight, windowHeight);

Setviewport 2
PC 4
GFX
OpenGL
Scaleform Flash
4.2
# 10 : DirectX
ShadowVolume OnFrameRender() D3D
BeginScene() EndScene()
EndScene() GfxTutorial::AdvanceAndRender()

```

```

void AdvanceAndRender(void)
{
    DWORD mtime = timeGetTime();
    float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
    MovieLastTime = mtime;

    pUIMovie->Advance(deltaTime, 0);
    pRenderer->BeginFrame();

    if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))

```

```

        {
            pRenderer->Display(hMovieDisplay);
        }

        pRenderer->EndFrame();
    }

Advance(deltaTime)

Gfx::Movie::Advance

# 11 :

Render::Renderer2D::Display DirectX D3D

D3D

Render::Renderer2D::Display D3D

Display
ScaleForm

DX9

DX9

GfxTutorial::AdvanceAndRender()

// GF DirectX
g_pStateBlock->Capture();

//
gFx->AdvanceAndRender();

// DirectX
g_pStateBlock->Apply();

# 12 : UI

DXUT UI

Section 4.1 \ShadowVolume.cpp

diff DXUT

// UI
***

DirectX Flash

```

図 3: G F x F l a s h U I S h a d o w V o l u m e

4.2 スケーリング モード

```
G F x :: M o v i e :: S e t V i e w p o r t
Flash
Scale form
:
X
X
4 : 3
Scale form
1
( 8 0 0 x 6 0 0 1 6 0 0 x 1 2 0 0 )
Scale form
```

```
G F x :: M o v i e :: S e t V i e w S c a l e M o d e
G F x T u t o r i a l :: I n i t G F x ( )
G F x :: M o v i e
```

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode 4
:

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

SetViewAlignment

SM_NoScale SM_ShowAll
SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment SetViewScaleMode SM_NoScale

Scaleform ActionScript

ActionScript d3d9guide.fla

C++ ActionScript

SetViewScaleMode SetViewAlignment ActionScript Stage (State.scaleMode Stage.align)

4.3 入カイベントの処理

Shadow Volume Scaleform Flash
Flash

7

Gfx::Movie::HandleEvent

4.3.1 マウス イベント

```
Shadow Volume      MsgProc      Win32
GfxTutorial::ProcessEvent      ScaleForm
WM_MOUSEMOVE WM_LBUTTONDOWN WM_LBUTTONUP

void ProcessEvent(HWND hWnd, UINT uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);
    if (pUIMovie)
    {
        if (uMsg == WM_MOUSEMOVE)
        {
            MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONDOWN)
        {
            ::SetCapture(hWnd);
            MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONUP)
        {
            ::ReleaseCapture();
            MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
    }
}

ScaleForm

# 1 :

pMovie->SetViewport(screen_width, screen_height, 0, 0,
                    screen_width, screen_height, 0);

( 0 , 0 )      Windows
ScaleForm
```

```
# 2 :
```

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,  
                    screen_width / 4, screen_height / 4, 0);
```

```
HandleEvent
```

```
Windows
```

```
Scaleform
```

```
# 3 :
```

```
movie_width = screen_width / 6;  
movie_height = screen_height / 6;  
pMovie->SetViewport(screen_width, screen_height,  
                    screen_width / 2 - movie_width / 2,  
                    screen_height / 2 - movie_height / 2,  
                    movie_width, movie_height);
```

```
Windows
```

```
(0, 0)
```

```
(screen_width / 2 - movie_width / 2, screen_height / 2 - movie_height / 2)
```

```
Windows
```

```
Flash
```

```
G F x :: M o v i e :: S e t V i e w A l i g n m e n t
```

```
G F x :: M o v i e :: S e t V i e w p o r t
```

```
SetViewAlignment
```

```
SetViewScaleMode
```

```
Scaleform
```

4.3.2 キーボード イベント

```
G F x :: M o v i e :: H a n d l e E v e n t
```

```
G F x :: K e y E v e n t
```

```
G F x :: C h a r E v e n t 2
```

```
KeyEvent(EventType eventType = None,  
          Key::Code code = Key::None,  
          UByte asciiCode = 0,  
          UInt32 wcharCode = 0,  
          UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

```
G F x :: K e y E v e n t
```

```
G F x :: C h a r E v e n t
```

```
A S C I I
```

```
Windows
```

```
G F x :: C h a r E v e n t
```

```
W M _ C H A R
```

```

G F x :: K e y E v e n t s      W M _ S Y S K E Y D O W N      W M _ S Y S K E Y U P
W M _ K E Y D O W N      W M _ K E Y U P
:

y  S h i f t      c      G F x :: K e y E v e n t      W M _ K E Y D O W N
:

f c
f
f S h i f t

y      W M _ C H A R      G F x :: C h a r E v e n t
A S C I I      'C'      S c a l e f o r m

y      c      W M _ K E Y U P      G F x :: K e y E v e n t
G F x :: C h a r E v e n t

y  F 5      G F x :: K e y E v e n t
2      F 5      A S C I I
G F x :: C h a r E v e n t

G F x :: K e y E v e n t
G F x _ E v e n t . h      F l a s h
G F x P l a y e r T i n y . c p p
S c a l e f o r m   P l a y e r      W i n d o w s      F l a s h
P r o c e s s K e y E v e n t
3 D

```

```

void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam,
                    LPARAM lParam)

```

```

W M _ C H A R      W M _ S Y S K E Y D O W N      W M _ S Y S K E Y U P      W M _ K E Y D O W N      W M _ K E Y U P
W i n d o w s W n d P r o c      S c a l e f o r m
p M o v i e

G F x :: K e y E v e n t s      G F x :: C h a r E v e n t s
A S C I I      G F x :: C h a r E v e n t s
U n i c o d e      (      [ I M E ]      )
I M E      I M E      (
I M E      )
P a g e U p      P a g e D o w n      G F x :: K e y E v e n t

```

Tutorial \ Section 4.3

3 D [S e t t i n g s] ()
C + + D X 9 d 3 d 9 g u i d e . f l a A c t i o n S c r i p t

[C h a n g e M e s h] ()

[C h a n g e M e s h] ()

F l a s h

4.3.3 ヒット テスト

3 D

U I 1

U I 3 D

G F x :: M o v i e :: H i t T e s t

F l a s h

G F x T u t o r i a l :: P r o c e s s E v e n t

H i t T e s t

U I

D X U T

```
bool processedMouseEvent = false;
if (uMsg == WM_MOUSEMOVE)
{
    MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONDOWN)
{
    ::SetCapture(hWnd);
    MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONUP)
{
    ::ReleaseCapture();
    MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
}
```

```

        processedMouseEvent = true;
    }

    if (processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
Movie::HitTest_Shapes))
        *pbNoFurtherProcessing = true;

```

4.3.4 キーボード フォーカス

```

[ C h a n g e M e s h ] (
                                4 . 3 . 2
                                W S A D Q E
                                3 D
                                S c a l e f o r m
                                3 D
                                6 . 1 . 2
                                F l a s h A c t i o n S c r i p t
                                C + +

```

4.3.5 タッチイベント

```

S c a l e f o r m
7
W i n d o w s A P I

```

WINVER=0x601

Windows API Windows 7

```

S h a d o w V o l u m e . c p p
# d e f i n e E N A B L E _ M U L T I T O U C H
# i n c l u d e < w i n d o w s . h >
RegisterTouchWindow
MsgProc WM_TOUCH

```

```

if(uMsg == WM_TOUCH)
{
    ProcessTouchEvent(pUIMovie, uMsg, wParam, lParam);
}

```

ProcessTouchEvent WM_TOUCH

GFx::TouchEvent HandleEvent

```
G Fx::TouchEvent( unsigned int id, float x, float y, float wcontact = 0, float hcontact = 0, bool primary = true, float pressure = 1.0f )
```

SWF

GFx

0

GFx::GestureEvent

MultitouchInterface

Scaleform

```
class FxPlayerMultitouchInterface : public MultitouchInterface
{
public:
    //
    virtual unsigned GetMaxTouchPoints() const { return 2; }

    //
    virtual uint32 GetSupportedGesturesMask() const { return 0; }

    //
    virtual bool SetMultitouchInputMode(MultitouchInputMode) { return true; }
};
```

SetMultitouchInterface

5 ローカライズとフォントについて

Font and Text Configuration Overview ()

5.1 フォントの概要と機能

Scaleform

Flash

TTF

Windows Linux

Scaleform

Developer Center

Documentation

:

<http://gameware.autodesk.com/scaleform/developer/?action=doc&lang=jp>

5000

Scaleform

Scaleform

1

Scaleform

TTF

Scaleform

Bin\Data\AS2\Samples\FontConfig

sample.swf

GFXPlayerD3D9

おはようございます。
お元気ですか？
さみしかったです。



図 4a:

図 4b:

C t r l W

2

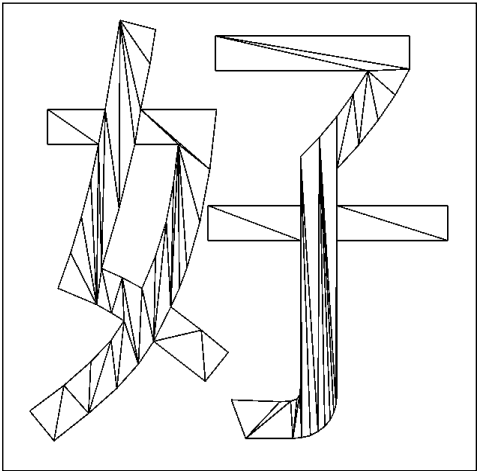


図 4c:

図 4d:

4 c 4 d S c a l e f o r m

F l a s h

F o n t O v e r v i e w (J a p a n e s e)

D e v e l o p e r C e n t e r

G F x _ 2 . 1 _ t e x t e f f e c t s _ s a m p l e . z i p

图 5

6 C++、Flash、ActionScript の相互作用

Flash ActionScript

ActionScript

Flash

ActionScript

Scaleform C++

ActionScript

ActionScript

Scaleform ActionScript C++

6.1 ActionScript から C++へ

Scaleform C++

FSCommand ExternalInterface 2

ActionScript

FSCommand ExternalInterface

C++

GFX::Loader

FSCommand

ActionScript fscommand

2

ExternalInterface

ActionScript

flash.external.ExternalInterface.call

GFX::Value (6.1.2))

FSCommand ExternalInterface

FSCommand

FSCommand

GFXExport -fstree -fslist -

fsparams

SWF

FSCommand

ExternalInterface

FSCommand

6.1.1 FSCommand のコールバック

ActionScript fscommand

ActionScript

:

fscommand("setMode", "2");

fscommand

ExternalInterface

G F x : : F S C o m m a n d H a n d l e r

```
function toggleFullscreen(ev:MouseEvent) {
```

```
    fscommand("ToggleFullscreen");
```

```
}
```

As an exercise, change `fscommand("ToggleFullscreen")` to

`fscommand("ToggleFullscreen", String(hud.visible))` to print the value of the `hud.visible` value to C++. Export the flash movie (CTRL+ALT+Shift+S) and replace `d3d9guideAS3.swf`.

```
fscommand("ToggleFullscreen")    fscommand("ToggleFullscreen", hud.visible)
```

```
hud.visible.value                C++                                Flash
```

```
(Ctrl+Alt+Shift+S) d3d9guide.swf
```

```
UI                                FSCommand
```

```
ShadowVolume                     fscommand
```

```
:
```

```
if (strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

```
DXUT                                OnFrameMove
```

```
OnFrameMove                        :
```

```
if (doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

```
Advance                            Invoke
```

6.1.2 ExternalInterface

```
Flash                                ExternalInterface.call    fscommand
```

```
ExternalInterface                fscommand
```

```
class OurExternalInterfaceHandler : public ExternalInterface
{
public:
    virtual void Callback(Movie* pmovieView,
```

```

        const char* methodName,
        const Value* args,
        unsigned argCount)
    {
        GFxPrintf("ExternalInterface: %s, %d args: ",
                    methodName, argCount);
        for(unsigned i = 0; i < argCount; i++)
        {
            switch(args[i].GetType())
            {
                case Value::VT_Null:
                    GFxPrintf("NULL");
                    break;
                case Value::VT_Boolean:
                    GFxPrintf("%s", args[i].GetBool() ? "true" : "false");
                    break;
                case Value::VT_Int:
                    GFxPrintf("%s", args[i].GetInt());
                    break;
                case Value::VT_Number:
                    GFxPrintf("%3.3f", args[i].GetNumber());
                    break;
                case Value::VT_String:
                    GFxPrintf("%s", args[i].GetString());
                    break;
                default:;
                    GFxPrintf("unknown");
                    break;
            }
            GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
        }
        GFxPrintf("\n");
    }
};

```

G F x :: L o a d e r

```

Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);

```

A c t i o n S c r i p t

O p e n d 3 d 9 H U D . a s u s e d b y

d 3 d 9 g u i d e A S 3 . f l a a n d l o o k a t t h e A c t i o n S c r i p t f o r t h e f o l l o w i n g f u n c t i o n s :

```

function onSubmit(path:String) {
    ExternalInterface.call("MeshPath", path);
    CloseMeshPath();
}

```

```

function onFocusIn(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", true);
}

function onFocusOut(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", false);
}

```

MeshPathFocus

ExternalInterface

ExternalInterface

(true

false)

"MeshPathFocus"

:

```

Callback! MeshPathFocus, nargs = 1
arg(0) = true

```

```

Callback! MeshPathFocus, nargs = 1
arg(0) = false

```

G F X T u t o r i a l

```

if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
args[0].GetType() == Value::VT_Boolean) {
    if (args[0].GetType() == Value::VT_Boolean)
        gfx->SetTextboxFocus(args[0].GetBool());
}

```

G F X T u t o r i a l :: P r o c e s s E v e n t

3 D

```

if (uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN || uMsg == WM_KEYUP ||
    uMsg == WM_CHAR)
{
    if (textboxHasFocus || wParam == 32 || wParam == 9)
    {
        ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
        *pbNoFurtherProcessing = true;
    }
}

```

```

        (ASCII code 32)          (ASCII code 9)          [Toggle UI] (UI
[Settings] (
                                [Change Mesh] (
                                Enter
Enter
                                ActionScript
                                ExternalInterface
OurExternalInterfaceHandler::Callback
:

static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
    doChangeMesh = true;
    const char *filename = args[0].GetString();
    MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
        changeMeshFilename, _countof(changeMeshFilename));
}

                                DXUT_OnFrameMove
DXUT

```

6.2 C++から ActionScript へ

```

6.1          ActionScript      C++
Scaleform
Scaleform      C++          ActionScript
ActionScript

```

6.2.1 ActionScript 変数の操作

```

Scaleform      GetVariable      SetVariable      2      ActionScript
Tutorials\Section 6.2
HUD          (fxplayer.swf)          FS

```

```

void GfxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM
    wParam, LPARAM lParam, bool *pbNoFurtherProcessing)
{

```

```

        int mx = LOWORD(lParam), my = HIWORD(lParam);

    if (pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if (wParam == VK_F5)
        {
            int counter = (int)pHUDMovie
                ->GetVariableDouble("_root.counter");

            counter++;
            pHUDMovie->SetVariable("_root.counter",
                Value((double)counter));

            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_root.MessageText.text", str);
        }
    }
}

```

...

```

GetVariableDouble _root.counter
GetVariableDouble SetVariable
_root.counter G F x :: M o v i e GetVariable
SetVariable

f x p l a y e r F l a s h 2
M e s s a g e T e x t
H U D T e x t _ r o o t . c o u n t e r
S e t V a r i a b l e

```

パフォーマンスの注意点:

```

Flash
Textfield.text G F x :: M o v i e :: I n v o k e A c t i o n S c r i p t
( 6 . 2 . 2 )
S c a l e f o r m

```

しないでください

8. HUD

```

SetVariable

GFX::Value

SetVariable          sticky (          )          GFX::Movie::SetVarType          3

Flash
          _root.mytextfield
3
SetVariable("_root.mytextfield.text", "testing", SV_Normal)          1
          SV_Sticky (          )
          _root.mytextfield.text          3
C++          SV_Normal

SV_Sticky          SV_Normal

SetVariableArray          Flash

GFXTutorial::InitGFX()          _root.SceneData

//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
                          "_root.SceneData", 0, sceneData, 3);

```

注意:

C++

C++

ActionScript

6.2

d3d9guide.swf/fla

ActionScript

6.2.2 ActionScript サブルーチンの実行

ActionScript

ActionScript

Gfx::Movie::Invoke

UI

UI

Gfx::Movie::Invoke

F6

[Change

Mesh] ()

F7

SetVariable

SetVariable

Invoke

ActionScript

Gfx::Movie::Invoke

WM_CHAR

openMeshPath

```

else if (wParam == VK_F6)
{
    bool retval = pHUDMovie->Invoke("root.ui.hud.OpenMeshPath", "");
    GfxPrintf("root.ui.hud.OpenMeshPath returns '%d'\n", (int) retval);
}
else if (wParam == VK_F7)
{
    const char *retval = pHUDMovie->Invoke(
        "root.ui.hud.CloseMeshPath", "");
    GfxPrintf("root.ui.hud.CloseMeshPath returns '%d'\n", (int) retval);
}

```

Invoke

1

ActionScript

Scaleform

ActionScript

1

ActionScript

```

      G F x :: M o v i e :: A d v a n c e
G F x :: M o v i e D e f :: C r e a t e I n s t a n c e      t r u e      i n i t F i r s t F r a m e

      I n v o k e      p r i n t f

G F x :: V a l u e      I n v o k e A r g s
                                v a _ l i s t      I n v o k e

I n v o k e      I n v o k e A r g s      p r i n t f      v p r i n t f

```

6.3 複数の Flash ファイル間の通信

```

C + +
U I
S W F

C + +

( M M O G )
H U D
H U D

H U D
H U D

3
S W F
C + +
3
3

G F x :: M o v i e

A c t i o n S c r i p t
l o a d M o v i e
u n l o a d M o v i e

2
S W F
G F x :: M o v i e

C + +

A c t i o n S c r i p t
c o n t a i n e r . s w f

/
c o n t a i n e r . s w f

/
A c t i o n S c r i p t

M o v i e C l i p L o a d e r

var mclLoader:MovieClipLoader = new MovieClipLoader();

A c t i o n S c r i p t
F l a s h

function loadMovie(url:String):void {
    var loader = new Loader();
    loader.load( new URLRequest( url ) );
    addChild( loader )
}

```

loadMovie

```
( _ r o o t . i n v e n t o r y W i n d o w . w i d g e t 1 . c o u n t e r )
```

```

Flash
    _root
        _root
            _root.counter
            _level6.counter
loadMovie
    _root.myMovie
        _root.counter

```

```

_lockroot = true
_lockroot
_root
ActionScript
    ActionScript
        AdobeFlash

```

G F x :: M o v i e

container.fla

(

)

```

loadMovie
    loadClip

```

(

) Loader

container.fla

ActionScript C + +

ExternalInterface

```

function loadMovie(url:String):void {
    var loader = new Loader();
    loader.load( new URLRequest( url ) );
    loader.contentLoaderInfo.addEventListener( Event.OPEN, handleLoadOpen,
                                                false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.INIT, handleLoadInit,
                                                false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.PROGRESS,
                                                handleLoadProgress, false, 0, true );
    loader.contentLoaderInfo.addEventListener( Event.COMPLETE,
                                                handleLoadComplete, false, 0, true );
    addChild( loader )
}

function handleLoadOpen( e:Event ):void {
    trace("Event.OPEN - Load Started!");
}

function handleLoadInit( e:Event ):void {
    trace("Event.INIT - Loaded Content Ready!");
}

```

```

function handleLoadProgress( e:Event ):void {
    trace("Event.PROGRESS - Load Progress!");
}
function handleLoadComplete( e:Event ):void {
    trace("Event.COMPLETE - Load Complete!");
}

```

7 テクスチャへのレンダリング

3 D

S W F

S W F

F l a s h S t u d i o

U I

U I

1 .

cell.x

4

cellwall.jpg

UI

cellwallRT.jpg

MeshMaterialList

```
Material {
    1.000000;1.000000;1.000000;1.000000;;
    40.000000;
    1.000000;1.000000;1.000000;;
    0.000000;0.000000;0.000000;;
    TextureFilename {
    "cellwallRT.jpg";
    }
}
```

2 .

cell.x

DXUT

DXUT

CreateTexture

Render

Textures

DXSDK

DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,
    D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
    D3DPPOOL_DEFAULT, &g_pRenderTex, 0);
```

```
g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3 .

Advance And Render

Gfx

a.

—

```
pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
```

b.

```
ptex->GetSurfaceLevel(0, &psurface);
HRESULT hr = pd3dDevice->SetRenderTarget(0, psurface );
```

c.

Display

a

8 OpenGL サンプル

```
N v i d i a                                O p e n G L / G L U T
(T u t o r i a l O p e n G L )
( h t t p : / / d e v e l o p e r . d o w n l o a d . n v i d i a . c o m / S D K / 1 0 / o p e n g l / s a m p l e s . h t m l # t e s s e l l a t i o n )
D 3 D 9
t e s s e l l a t i o n . c p p      t e s s e l l a t i o n _ o r i g i n a l . c p p
N V I D I A
G F x      I n i t ( )      S h u t D o w n      A d v a n c e A n d D i s p l a y ( )
G F x      G F x P l a y e r I m p l
( G F x P l a y e r G L . c p p )
```

9 GFxExport を使った前処理

```
F l a s h      S W F
S W F
S W F
S W F
S W F
S W F
D D S
D D S
G F x E x p o r t
G F x E x p o r t
G F x
G F x E x p o r t
d 3 d 9 g u i d e . s w f      f x p l a y e r . s w f
S c a l e f o r m      :
```

```
gfxexport -i DDS -c d3d9guideAS3.swf
gfxexport -i DDS -c fxplayer.swf
```

```
G F x e x p o r t . e x e      C : \ P r o g r a m   F i l e s \ S c a l e f o r m \ $ ( G F X S D K )
T u t o r i a l \ S e c t i o n 7      c o n v e r t . b a t
```

```
- i      D D S      D D S      D i r e c t X
D X T
4
```

- c

S c a l e f o r m

A c t i o n S c r i p t

D X T

- s h a r e _ i m a g e s

S W F

1

T u t o r i a l S e c t i o n 8

S h a d o w V o l u m e

G F x :: L o a d e r :: C r e a t e M o v i e

S c a l e f o r m

10 次の段階へ

	Scaleform	
	Flash	
X	Scaleform Player SWF to Texture SDK	Gamebryo™
	Unreal Engine 3	
X	Developer Center	FAQs
X	Scale9	Documentation
	Overview	Scale9 Grid
X	Scaleform SWF	Flash
		Gfx::File Opener
X	Gfx::Image Creator	
X	Gfx::Translator	